

Lab Explanation-05

1(a)

Given that I must use DFS. Therefore, using the reverse-list, I first took the vertex of those who do not have a degree. I then applied dfs to those. I created a 2D list called timer and recorded the beginning and ending times of each vertex.

I determined whether there was a cycle or something else by comparing the length of the visited list to the vertex list, so I wrote the output.

If all goes well, I sort the vertex list in a dictionary called b by rearranging it in accordance with the end time desc order. At last, I processed the dictionary result.

1(b)

I've used BFS here. To start, I created a list called in_degree_counter that contains the degree of freedom that each vertex has.

I began by appending to a queue the vertex whose in-degree is zero. And as it emerges, it adds itself to the visited list by becoming the nearest vertex in-degree to (their in-degree-1).

In a similar manner, the loop was repeated to check if any vertices with a degree of zero were not on the visited list. Ultimately, the data from the visited list was processed.

2

It is the similar logic of the 1(a). Here we just modify the pre_selection to the reverse sorted form so the final result the lowest comes first and the highest comes last.

3

Here, the cycle and its path are being detected. Therefore, it will be determined whether or not each vertex that has not been visited is in a cycle. If not, the information will be printed normally; if not, a temporary visited list will be created to help locate the path.